

## EXPT NO -3 NAND and NOR gates as Universal gates.

### THEORY:

NAND and NOR gate can used to construct and perform the basic funtions of all the basic logic gates and hence they are known as Universal gates.

#### (a) NAND as NOT gate:

We know that the NOT gate acts as a inverter i.e the output of the NOT gate is in the complement form of the input. We know that there are two inputs for a NAND gate and for a NOT gate there is only input. So to construct a NOT gate from the NAND gate we need to modify the Boolean expression of a NOT gate. The Boolean expression is shown th expression (1)

$$Y = \overline{A} \quad \dots\dots\dots (1)$$

The expression can also be written as

$$Y = \overline{A \cdot A} \quad \dots\dots\dots (2) \text{ \{ Since } A = A \cdot A \text{ \}}$$

Therefore inorder to construct a NOT gate from a NAND gate the inputs of the NAND gate are shorted to form a single input.

The logic circuit diagram is shown below in figure (3.1)

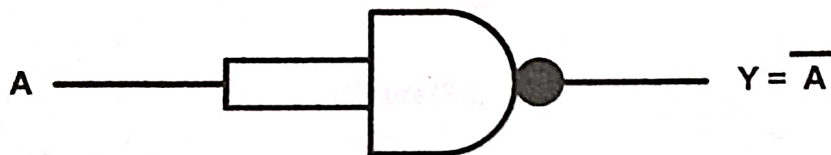


figure (3.1)

#### (b) NAND as AND gate:

The Boolean Expression of a AND gate is given the expression (3).

$$Y = A \cdot B \quad \dots\dots\dots (3)$$

The above expression can be written as  $\overline{\overline{A \cdot B}}$

$$Y = A \cdot B \quad \dots\dots\dots (4) \text{ \{ Double negation \}}$$

Hence we see from the above expression that we require two NAND gates to construct a AND gate. The logic circuit diagram is shown below in figure (3.2)

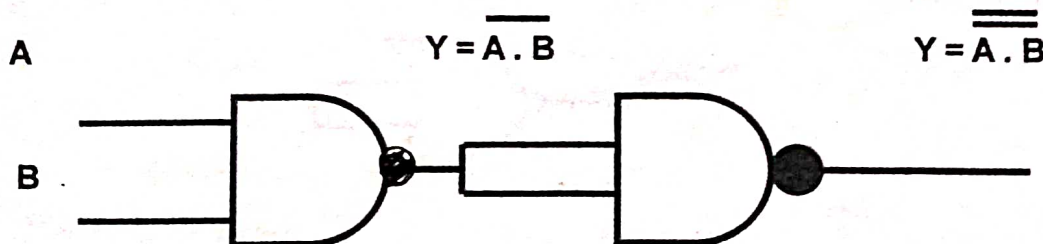


Figure (3.2)

(c) NAND as OR gate:

The Boolean Expression of a OR gate is given the expression (5).

$$Y = A + B \quad \dots\dots\dots (5)$$

The above expression can be written as

$$Y = \overline{\overline{A + B}} \quad \dots\dots\dots (6) \text{ \{ Double negation \}}$$

$$Y = \overline{\overline{A}} \cdot \overline{\overline{B}} \quad \dots\dots\dots (7) \text{ \{ Applying De - Morgans Theorm \}}$$

Hence, we see from the above expression that we require three NAND gates to construct a OR gate.  
The logic circuit diagram is shown below in figure (3.3)

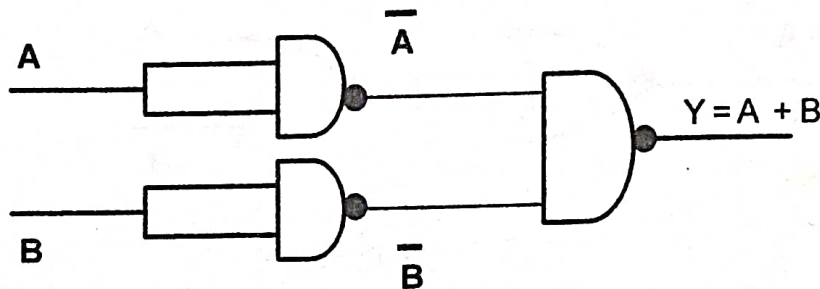


Figure (3.3)

(d) NOR as NOT gate:

We know that the NOT gate acts as a inverter i.e the output of the NOT gate is in the complement form of the input. We know that there are two inputs for a NOR gate and for a NOT gate there is only input. So to construct a NOT gate from the NOR gate we need to modify the Boolean expression of a NOT gate. The Boolean expression is shown th expression (1)

$$Y = \overline{A} \quad \dots\dots\dots (8)$$

The expression can also be written as

$$Y = \overline{A + A} \quad \dots\dots\dots (9) \text{ \{ Since } A = A + A \text{ \}}$$

Therefore inorder to construct a NOT gate from a NOR gate the inputs of the NOR gate are shorted to form a single input.

The logic circuit diagram is shown below in figure (3.4)



figure (3.4)

(e) NOR as OR gate:

The Boolean Expression of a OR gate is given the expression (10).

$$Y = A + B \quad \dots\dots\dots (10)$$

The above expression can be written as

$$\overline{\overline{Y}} = \overline{\overline{A + B}} \quad \dots\dots\dots (11) \text{ \{ Double negation \}}$$

Hence, we see from the above expression that we require two NOR gates to construct a OR gate. The logic circuit diagram is shown below in figure (3.5)

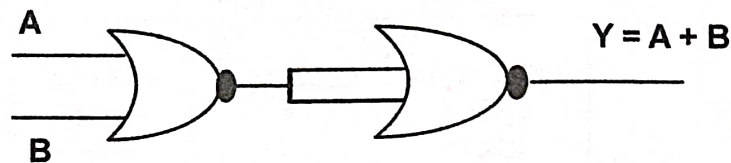


Figure (3.5)

(f) NOR as AND gate:

The Boolean Expression of a AND gate is given the expression (5).

$$Y = A \cdot B \quad \dots\dots\dots (5)$$

The above expression can be written as

$$\overline{\overline{Y}} = \overline{\overline{A \cdot B}} \quad \dots\dots\dots (6) \text{ \{ Double negation \}}$$

$$\overline{\overline{Y}} = \overline{\overline{A} + \overline{B}} \quad \dots\dots\dots (7) \text{ \{ Applying De - Morgans Theorem \}}$$

Hence, we see from the above expression that we require three NOR gates to construct a AND gate. The logic circuit diagram is shown below in figure (3.6).

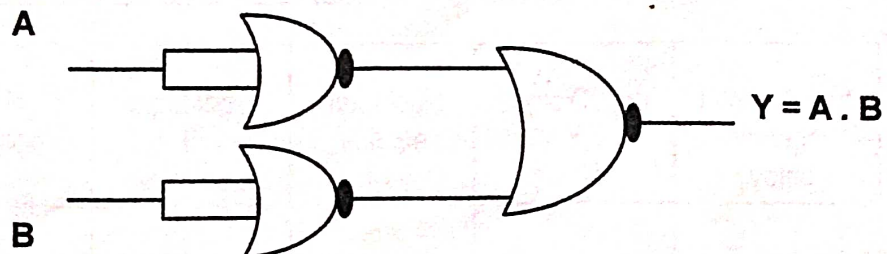


figure (3.6).

### APPARATUS AND DEVICES REQUIRED:

(a) Semiconductor IC

(i) Quad Two input NAND gate : 74LS00

(ii) Quad Two input NOR gate : 74LS02

(b) Jumpers

(c) Bread Board

(d) + 5 Volts Regulated Power supply.

(e) Light emitting Diode (LED)

### EXPERIMENTAL OBSERVATION:

(a) Truth Table of NOT gates constructed out of NAND gate:

| Input logic<br>A | Input logic<br>Voltage<br>(Volts) | Logic output<br>Y | Logic output<br>Voltage<br>(Volts) |
|------------------|-----------------------------------|-------------------|------------------------------------|
|                  |                                   |                   |                                    |

(b) Truth Table of OR gates constructed out of NAND gate:

| Input logic<br>A | Input logic<br>Voltage | Input logic<br>B | Input logic<br>Voltage | Logic Output<br>Y | Logic Output<br>Voltage<br>(Volts) |
|------------------|------------------------|------------------|------------------------|-------------------|------------------------------------|
|                  |                        |                  |                        |                   |                                    |

(c) Truth Table of AND gates constructed out of NAND gate:

| Input logic<br>A | Input logic<br>Voltage | Input logic<br>B | Input logic<br>Voltage | Logic Output<br>Y | Logic Output<br>Voltage<br>(Volts) |
|------------------|------------------------|------------------|------------------------|-------------------|------------------------------------|
|                  |                        |                  |                        |                   |                                    |

(d) Truth Table of NOT gates constructed out of NOR gate:

| Input logic<br>A | Input logic<br>Voltage<br>(Volts) | Logic output<br>Y | Logic output<br>Voltage<br>(Volts) |
|------------------|-----------------------------------|-------------------|------------------------------------|
|                  |                                   |                   |                                    |

(e) Truth Table of OR gates constructed out of NOR gate:

| Input logic<br>A | Input logic<br>Voltage | Input logic<br>B | Input logic<br>Voltage | Logic Output<br>Y | Logic Output<br>Voltage<br>(Volts) |
|------------------|------------------------|------------------|------------------------|-------------------|------------------------------------|
|                  |                        |                  |                        |                   |                                    |

(f) Truth Table of AND gates constructed out of NOR gate:

| Input logic<br>A | Input logic<br>Voltage | Input logic<br>B | Input logic<br>Voltage | Logic Output<br>Y | Logic Output<br>Voltage<br>(Volts) |
|------------------|------------------------|------------------|------------------------|-------------------|------------------------------------|
|                  |                        |                  |                        |                   |                                    |

### DISCUSSION:

- (i) The bread makes circuit connections highly flexible. There is no need for soldering or binding screws.
- (ii) The light emitting diode must be connected using proper current limiting diode to avoid burn outs.
- (iii) The power supply must be connected to the supply pin of the TTL chips with proper polarity and proper voltage rating.
- (iv) The TTL chips connected on the bread board must be properly fixed in order to get better connection and results.
- (v) The power supply connected must be positive five volts regulated power supply. The voltage must never exceed the tolerable limit.